



Review article

A comprehensive review of Artificial Intelligence regulation: Weighing ethical principles and innovation

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ARTICLE INFO

Keywords:

Artificial Intelligence

Ethical principles

Innovation

Regulation

Risks

ABSTRACT

We present a comprehensive review of Artificial Intelligence (AI) regulation, addressing the challenges and needs associated with governing rapidly evolving AI technologies. We begin by exploring the definition of AI, the risks associated with AI algorithms, and possible approaches to address them. This foundation sets the stage for a detailed discussion on AI regulation across several dimensions, including principles of ethical AI, specific strategies for regulating AI, and the challenges related to AI regulation. We conclude by offering insights to help regulators and society find the most balanced approach to AI regulation, and by proposing key questions that should be considered during the development of AI regulations. Our presentation explores the multifaceted nature of AI regulation, emphasizing the importance of integrating insights from computer science, law, and economics to develop robust regulatory frameworks.

1. Introduction

Artificial intelligence (AI) has significantly influenced virtually every dimension of contemporary life, transforming economies, institutions, and everyday activities. While AI has primarily served as a productivity-enhancing tool—helping individuals and organizations make more efficient decisions—it has also introduced critical risks, such as privacy violations, the amplification of existing biases in data, and the development of AI systems that are difficult to interpret. Notable incidents have highlighted these concerns. For example, researchers identified individual Netflix users by correlating anonymized recommendation data with publicly available movie ratings, exposing serious vulnerabilities in personal data protection (Narayanan and Shmatikov, 2006). Similarly, the COMPAS algorithm, extensively used within the U.S. judicial system, faced significant criticism because it appeared to disproportionately affect Black defendants, raising profound ethical and fairness concerns about AI-driven decision-making (Christian, 2020). The controversy surrounding Ofqual's algorithmic grading system during the COVID-19 pandemic in the UK also underscored severe transparency issues in AI systems, significantly affecting trust in technology-enabled assessments (Katwala, 2020). These incidents collectively emphasize the urgent need for robust and effective AI regulation to maximize societal benefits while proactively addressing and mitigating associated risks.

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<https://doi.org/10.1016/j.ject.2025.07.001>

Received 30 March 2025; Received in revised form 9 July 2025; Accepted 9 July 2025

Available online 11 July 2025

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Given this context, our paper aims to provide a comprehensive and multidisciplinary review of AI regulation, integrating insights from economics, computer science, and law. We explore central questions about how we can and should regulate AI, addressing the inherent challenges in defining AI, managing algorithmic risks, embedding ethical principles, and choosing among different regulatory strategies. Crucially, we emphasize that effective AI governance must balance innovation and economic competitiveness with societal safeguards, ethical standards, and legal accountability. This paper serves as a guide for policymakers, researchers, and practitioners by systematically clarifying key concepts, synthesizing relevant findings from multiple disciplines, and outlining viable regulatory frameworks to address the nuanced challenges posed by AI.

Reviewing the literature on AI regulation brings several challenges. First, many papers are available, raising an important question: Which ones should we include? Second, these papers often present very different views and methods, leading to another key question: How can we clearly organize and present these differences? Before we continue, we want to clarify that our review does not focus on specific laws or regulations that countries have already adopted or plan to adopt. Instead, we focus on broader ideas, such as how to manage risks, address ethical questions, and choose among different regulatory options. To select papers, we use a citation-based approach. We start with influential and frequently cited works in the AI field. From these papers, we follow both backward citations (older important papers cited by the ones we chose) and forward citations (newer papers that cite the ones we chose).

Several relevant works addressing the issue of AI regulation have naturally influenced our own. In particular, we cite [Scherer \(2015\)](#); [Hoffmann-Riem \(2020\)](#); and [Lobel \(2023\)](#) as fundamental reviews on the topic. Beyond these core reviews, a broader set of interdisciplinary works has significantly influenced discussions on AI regulation. In particular, substantial contributions from computer scientists have advanced technical solutions to improve fairness, privacy, and interpretability in AI systems, as exemplified by [Kearns and Roth \(2019\)](#) and [Molnar \(2020\)](#). Economists, meanwhile, have emphasized the impact of regulatory frameworks on innovation dynamics and market entry barriers, noting the potential economic consequences of overly restrictive or overly permissive regulatory environments ([Klapper et al., 2006](#); [Palagashvili and Suarez, 2024](#)). Legal scholars have extensively explored liability issues, debating the optimal approaches for assigning responsibility in cases of harm caused by AI systems ([Zech, 2021](#)). Our paper differentiates itself by explicitly integrating these distinct disciplinary perspectives, providing a unified and nuanced analysis of AI regulation grounded in economic theory, legal principles, and computational insights.

The remainder of the paper proceeds as follows. [Section 2](#) discusses the definitional challenges associated with AI and its various subfields, with particular emphasis on Machine Learning (ML) as the predominant focus of regulatory concerns. [Section 3](#) explores specific algorithmic risks related to privacy violations, bias from flawed datasets, and the lack of interpretability, reviewing both real-world cases and potential technical solutions. [Section 4](#) outlines critical ethical principles, such as justice, transparency, and accountability, that should guide the responsible development and deployment of AI technologies. [Section 5](#) systematically compares regulatory strategies, including self-regulation, risk-based command-and-control models, and innovation-friendly regulatory sandboxes. It also discusses sector-specific versus broad regulatory frameworks and strict versus subjective liability approaches. Finally, [Section 6](#) addresses practical challenges to implementing AI regulations effectively, and [Section 7](#) concludes by providing structured recommendations and posing key questions for policymakers and stakeholders involved in shaping AI regulatory landscapes.

2. Artificial Intelligence (AI)

In this section, we introduce AI and divide the content into three subsections. In Subsection [2.1](#), we explain what AI means and discuss the challenges of defining it. In Subsection [2.2](#), we present the various fields of AI and highlight that concerns about AI use mainly related to the field of ML. Given its particular importance, we examine ML in more detail in Subsection [2.3](#).

2.1. What is AI?

Defining and understanding AI is crucial for specifying the scope of legislation concerning its use. Therefore, we begin by emphasizing the challenges involved in defining the term AI. This difficulty arises from the broad nature of the concept of intelligence (not limited to AI) and its applications, which complicates the task of specifying explicit characteristics.

The well-known definition of AI, proposed by John McCarthy, one of the founders of the discipline and the person who originally coined the term, describes AI as “the science and engineering of making intelligent machines, especially intelligent computer programs” ([McCarthy, 2007](#)). This definition hinges on the concept of *intelligence*, which McCarthy defines as “the ability to achieve goals in the world”. From a legal standpoint, however, this definition proves inadequate because it relies on an overly generic notion of “achieving goals in the world”.

Another common approach to defining AI is based on our current understanding of computational systems. In this context, we may consider a definition such as: “AI is the study of how to make computers do things at which, at the moment, people are better” ([Rich, 1983](#)). This definition associates intelligence with tasks that humans currently perform better than computers. However, it is important to recognize that what constitutes a complex task for a machine evolves over time. For example, in the 1980s, programs that emulated human chess playing had limited performance, making this activity a complex task for machines. Today, however, several systems play chess at levels that exceed most humans.

[Russell and Norvig \(2020\)](#) address the problem of defining AI along two dimensions: thought processes and behavior, and explore these dimensions in terms of human and rational performance. We summarize these dimensions in [Table 1](#).

[Martinez \(2018\)](#) adds the important element of *autonomy* to the definition of AI: “artificial intelligence is a system, program, software, or algorithm that acts autonomously to think rationally, think humanely, act rationally, act humanely, make decisions, or provide outputs”. The challenge in incorporating this new element lies in defining what constitutes an *autonomous* system. The

Table 1
Four dimensions for defining AI.

Dimension	Description	Typical Applications
Human thought process	Machines that imitate human cognitive processes.	Neural networks and deep learning systems.
Human behavior	Machines that behave indistinguishably from humans in interactions. We can connect this definition to the so-called ‘Turing’s imitation game’.	Chatbots and virtual assistants.
Rational thought process	Machines using logical reasoning and deduction to make decisions.	Expert systems and logic-based reasoning.
Rational behavior	Machines acting optimally based on goals, sensing, and adapting to their environment. This definition connects to the idea that intelligent machines behave by ‘doing the right thing’.	Reinforcement learning and agent-based systems.

concept of autonomy can vary depending on the level of independence a system exhibits. For example, we can consider an air conditioning system autonomous because it automatically increases cold air output as the room it aims to cool warms up. On the other hand, AlphaGo Zero teaches itself entirely and even outperforms the original AlphaGo, which famously defeated the world champion of Go.

There is no universally accepted definition of AI, which poses a fundamental problem for any proposal of general AI regulation (Ruscheimer, 2023). If we cannot clearly define what AI is, how can we effectively determine the scope of regulation?

2.2. The subfields of AI

Although defining AI precisely is inherently challenging, we can broadly divide it into three important subfields: ML, expert systems, and search algorithms. ML involves developing algorithms that enable computers to learn from data and improve their performance over time without being explicitly programmed for each specific task. Expert systems are computer programs designed to simulate the judgment and behavior of a human or group of humans with expertise in a specific area, using predefined rules. Search algorithms are computational methods that systematically explore available options in a problem to find a desired solution. While some researchers may consider robotics, Natural Language Processing (NLP), and computer vision as additional subfields of AI, due to their applied nature, we prefer to view them as applications of AI, or more specifically, of ML.

In terms of regulation, we concentrate most concerns on the field of ML due to its potential to generate unforeseen outcomes. While we design expert systems to simulate the decision-making abilities of human experts, they operate based on a set of predefined rules and a knowledge base. Thus, we limit them to the rules and knowledge encoded by human experts, and they lack the ability to adapt or learn beyond their initial programming. Similarly, search algorithms systematically explore a vast space of possible solutions to find the best outcome. While this process is automatic, the algorithm does not learn or adapt beyond its predefined search strategy; its behavior remains confined to efficiently performing the search within the established rules. In contrast, ML algorithms exhibit a higher degree of unpredictability. These algorithms can learn from data, adapt to new information, and improve their performance over time without explicit reprogramming. This type of behavior is more dynamic, allowing the algorithms to evolve with changing environments and data inputs, which makes their outcomes less predictable and therefore more challenging to regulate. Note that this unforeseeable behavior raises concerns not only about the potential loss of control over machines guided by ML algorithms but also about the difficulty of assigning moral and legal responsibility in the case of unexpected outcomes (Scherer, 2015). Since we cannot predict these outcomes, it becomes difficult to attribute responsibility to developers, companies that commercialize the systems, or users.

2.3. Machine learning (ML)

As we defined earlier, ML involves the development of algorithms and statistical models that enable computers to perform specific tasks without using explicit instructions, instead learning from data and making predictions or decisions based on it. In ML, we categorize learning algorithms based on how they process and learn from data to achieve specific objectives.

In *supervised learning*, the objective is to develop a model, denoted as $g(\theta, x)$, that approximates an unknown function f such that $y = f(x)$. Here, $\theta \in \Theta$ represents a vector of parameters within the set of admissible parameters Θ , and x is the input data. We aim to achieve this approximation by training the model using input data x and their corresponding target outputs y . We usually divide supervised models into *regression* models, when the target output is a continuous variable, and *classification* models, when the target output is a discrete variable.

Although we can use other performance indexes, the estimation of a regression model usually attempts to solve an optimization problem, where we choose a set of parameters $\theta \in \Theta$ that minimizes the squared error function, as in

$$\min_{\theta \in \Theta} \sum_{i=1}^n (y_i - g(x_i, \theta))^2, \quad (1)$$

where y_i is the target output and $x_i = [x_{i1}, \dots, x_{iK}]'$ is the vector of attributes associated with the i -th observation in a sample of n observations. The simplest case happens when $g(x_i, \theta) = \theta'x_i = \theta_1 x_{i1} + \dots + \theta_K x_{iK}$ is a linear model that depends on K parameters,

where $\theta = [\theta_1, \dots, \theta_K]'$ is the vector of parameters. In many cases, $x_{i1} = 1$, making θ_1 the intercept term. In this scenario, we can find the parameters of $g(X, \theta)$ by setting the derivative of the loss function, presented in Eq. (1), with respect to θ equal to zero, allowing us to obtain a closed-form solution for the parameters that minimize the loss function. Although for more complex models, such as neural networks, we cannot find a closed-form solution for the parameters that minimize the mean squared error, we can determine them numerically using variations of gradient methods.

We typically categorize classification models into two subgroups: binary classification and multinomial classification. In binary classification, the model distinguishes between two classes. In contrast, multinomial classification involves more than two classes.

In the case of the binary logit model, which is the simplest classification scenario, we assume that the random variable y_i , corresponding to observation i , follows a Bernoulli distribution.¹ We associate this target variable y_i with a vector of K attributes $x_i = [x_{i1}, x_{i2}, \dots, x_{iK}]'$. The probability of success, $P(y_i = 1|x_i)$, depends on the characteristics numerically represented by the components of the vector x_i . The goal of binary response models is to explain the probability of outcomes $y_i = 1$ or $y_i = 0$ using these explanatory variables. In the logit model, the probability model for y_i given x_i is expressed as

$$p(x_i) = P(y_i = 1|x_i) = F(x'\theta),$$

where F denotes the logistic distribution²

$$F(z) = \frac{\exp(z)}{1 + \exp(z)},$$

ensuring that the function $F(\cdot)$ is bounded within the interval $0 < F(z) < 1$ for all z in $\mathbb{R}$. We can estimate this model using the *maximum likelihood* method, which is a statistical technique employed to estimate the parameters of a probability distribution by maximizing a likelihood function, where the likelihood function measures how likely it is that a model, with its current parameter values, would produce the observed data. Given that y_i is a Bernoulli random variable with a mean conditioned on x_i , the density function for y_i given x_i is

$$f(y_i|x_i) = F(x'\theta)^{y_i}(1 - F(x'\theta))^{1-y_i}, \quad \forall y_i \in \{0, 1\}.$$

Assuming independence among observations, the likelihood function for the entire dataset is

$$L(X/\theta) = \prod_{i=1}^n f(y_i/x_i) = \prod_{i=1}^n [F(x'\theta)^{y_i}(1 - F(x'\theta))^{1-y_i}].$$

The corresponding log-likelihood function is thus³

$$\log L(X/\theta) = \sum_{i=1}^n \log f(y_i/x_i) = \sum_{i=1}^n [y_i \log F(x'\theta) + (1 - y_i) \log(1 - F(x'\theta))].$$

To estimate the model parameters effectively, we solve the maximization problem:

$$\max_{\theta} \log L(X/\theta). \quad (2)$$

This maximization leads to the most likely estimates of θ under the binary logit model, improving our understanding of the relationship between the predictors and the binary outcome. We may extend this approach for finding the vector of parameters θ associated with the binary logit model to the multinomial case and to more complex models, such as neural networks.

Classical applications of regression models include stock price prediction and energy consumption forecasting. On the other hand, classical applications of classification models include email spam detection, credit scoring, and default prediction.

In *unsupervised learning*, we only have access to the input data X without any explicit labels. The objective is to reorganize or interpret the data X based on inherent patterns or structures without reference to known, labeled outcomes, usually by an optimization criterion or a rule defined by the algorithm. Unsupervised ML models cover a wide variety of applications due to their diversity. These models are particularly effective in anomaly detection, including the identification of fraudulent activities and the detection of system faults, dimension reduction and graphical representation. A special class of unsupervised models are the models used for NLP, such as the vector space models and the Latent Dirichlet Allocation (LDA) models. A subset of unsupervised models includes those that aim to predict the next word in a sequence based on previous words, such as the embedding models that represent words in high dimensional Euclidean spaces. We say that these models are semi-supervised (or self-supervised) because, although they do not have explicit target outputs like traditional supervised models, they use large amounts of text data (such as text available on the internet) where each sequence of words serves as an implicit target for predicting the next word.

In *reinforcement learning*, the algorithm interacts with an environment and receives feedback in the form of rewards or penalties in response to the actions it takes. The data consists of responses from the environment to the actions performed by the algorithm. The

¹ A Bernoulli distribution is a discrete probability distribution where the Bernoulli random variable can only take on the values 0 or 1.

² This approach makes sense because we want the model's output to represent probabilities. Since the range of a cumulative probability distribution lies within the interval [0,1], this outcome occurs naturally.

³ Because logarithms are strictly increasing functions, maximizing the likelihood is equivalent to maximizing the log-likelihood. Thus, we apply the logarithm function to the likelihood of the entire dataset to simplify the optimization problem.

principal objective is to learn a policy, i.e., build a “table” that maps states (or situations) to actions, that maximizes the cumulative reward over time. Classical applications of reinforcement learning include gaming, robotics and dynamic pricing.

In general, there has been much discussion about the role of ML algorithms in the unforeseen behaviors that AI systems can exhibit. As the discussion of learning types highlights, ML algorithms consist of precise and systematic sequences of steps to estimate parameters or solve problems based on data (Kroll, 2018). However, since the data directly influences the model’s structure and learned parameters, the model may reflect the dataset’s characteristics and biases.

3. Algorithmic risks of AI

Like the widespread adoption of any new technology, AI brings both significant positive impacts and risks that we must understand and address. Three types of risks stand out (Agarwal and Mishra, 2021): the loss of data privacy, the perpetuation of biases present in the data, and the lack of interpretability or explainability of the models.

3.1. Privacy violations

Regarding data privacy loss, there are two specific risks. The first concerns the data itself: once collected, it can compromise the privacy of individuals whose characteristics it includes. The second risk arises after estimating AI models. In some cases, it is possible to recover individuals’ data from the predictions made by already estimated models.

In the mid-1990s, the Massachusetts Group Insurance Commission (GIC) decided to release hospital visit records of state employees to researchers. They assumed that removing explicit identifiers, such as name, address, and social security number, would protect patient privacy, despite leaving nearly one hundred other attributes per patient, including ZIP code, birth date, and sex. Governor Weld, who served as the 68th Governor of Massachusetts from 1991 to 1997, assured the public that he would protect patient privacy. However, Dr. Latanya Sweeney, then a graduate student, set out to find Weld’s hospital records in the GIC data. Knowing that Weld lived in Cambridge, Massachusetts, she purchased the city’s voter rolls for twenty dollars. This database included voters’ names, addresses, ZIP codes, birth dates, and genders. By combining the voter data with the GIC records, Sweeney quickly identified Weld. Only six people in Cambridge shared his birth date; only three of them were men; and of those, only Weld lived in his specific ZIP code. To highlight the vulnerability, Dr. Sweeney sent Weld’s health records, including diagnoses and prescriptions, to his office. In her paper, Sweeney (2000) showed that even after removing explicit identifiers like names and addresses, we can uniquely identify a high proportion of individuals in geographically situated populations based on a few demographic characteristics, such as 5-digit ZIP code, gender, and date of birth.

Another interesting example comes from the Kaggle Netflix competition. On October 2, 2006, Netflix, the world’s largest online movie rental service, publicly released 100 million records detailing how nearly half a million users rated movies from December 1999 to December 2005. Netflix anonymized these records by removing identifying information such as usernames, while assigning a unique user identifier to maintain continuity of ratings. Netflix aimed to improve its movie recommendation algorithm by launching the Netflix Prize, offering one million dollars to the team that could significantly improve its algorithm using the released data. Many researchers celebrated this initiative, as it allowed them to use the Netflix data to improve or create new recommendation engines. However, two weeks after the data release, Arvind Narayanan and Professor Vitaly Shmatikov from the University of Texas demonstrated that an attacker with minimal knowledge of a user’s movie-watching preferences could easily identify that user in the Netflix Prize dataset (Narayanan and Shmatikov, 2006). Using the IMDb database as an auxiliary dataset, they correlated ratings and timestamps from the Netflix data with publicly shared ratings and timestamps on IMDb (where users often do not anonymize their profiles). This approach allowed them to find patterns and overlaps that pointed back to specific users in the Netflix dataset. Their research revealed that knowing the exact ratings a user gave to six obscure movies could identify the user 84 % of the time. If the attacker knew approximately when a user rated six movies, they could identify the user 99 % of the time. Additionally, knowing the precise ratings and dates (within three days) of just two movies allowed the identification of 68 % of the users. Further, Narayanan and Shmatikov (2008) introduced a new class of statistical de-anonymization attacks targeting high-dimensional micro-data, such as individual preferences, recommendations, and transaction records, and applied these algorithms successfully to the Netflix dataset.

Furthermore, an interesting example from the administration of warfarin dosing shows that even after removing identifying data, we can still extract sensitive information from already estimated models. Warfarin dosing is critical in the field of pharmacogenetics due to the drug’s widespread use as an anticoagulant to prevent strokes in patients with atrial fibrillation and other cardiovascular conditions. The dosing of warfarin is particularly challenging because the appropriate dosage varies significantly among patients, and incorrect dosing can cause serious adverse events such as stroke or uncontrolled bleeding. Genetic variability plays a significant role in determining warfarin dosage requirements. Pharmacogenetic models that incorporate genetic markers, demographics, and clinical history can predict proper dosages more accurately. Fredrikson et al. (2014) showed that it is possible to invert these models and predict individuals’ genetic markers with significantly greater accuracy than random guessing.

The critical issue addressed here is how to extract useful information about the underlying population, as represented by the database, while maintaining the privacy of individuals (Dwork, 2006). First, it is essential to define what constitutes privacy. Second, it is necessary to identify what auxiliary information is available that one may use to retrieve sensitive information about individuals. A natural approach to defining privacy in data analysis is to ensure that an analyst’s views about an individual, both before and after accessing the database, do not significantly change. In other words, access to the database should not substantially alter their perception of any individual. Auxiliary information refers to any additional data outside the database that an analyst may use to infer sensitive details about individuals. This may include publicly available data such as census records, social media profiles, news articles, or even specific medical studies.

Table 2
Techniques to mitigate privacy violations.

Anonymization	Removes personally identifiable information from datasets to prevent re-identification, though studies show it is often ineffective due to auxiliary data.
k-Anonymity	Ensures each record is indistinguishable from at least $k - 1$ others by generalizing or suppressing quasi-identifiers, reducing re-identification risk.
Differential Privacy	Adds calibrated noise to data or outputs to prevent inference about any individual, offering strong, quantifiable privacy guarantees.
Federated Learning	Trains models locally on user devices and aggregates updates centrally, avoiding direct sharing of raw personal data.
Secure Multi-Party Computation (MPC)	Allows multiple parties to jointly compute results over private data without revealing their individual inputs.
Homomorphic Encryption	Enables computations directly on encrypted data, allowing secure outsourcing of data processing without revealing plaintext.

Based on these definitions, a technique we can use to protect individuals from privacy failure is known as differential privacy. It works by adding a small amount of random noise to the data or to the results of queries on the data. Using this approach, it becomes very difficult for anyone to determine whether the dataset includes a specific individual’s information, because the noise obscures any one person’s data without significantly altering the overall results or statistics derived from the dataset (Dwork, 2006). For instance, consider a survey conducted at a university to determine the proportion of students using illicit drugs. We may develop a website where students respond, using their registration number, whether they “regularly use any type of illicit drug”. An algorithm may record the actual response 80 % of the time and randomize it 20 % of the time. Thus, even if attackers compromise the database, they cannot confirm the authenticity of specific responses. However, knowing the pattern of introduced noise, we can statistically estimate the proportion of drug-using students (Kearns and Roth, 2019). Although this method requires a larger sample size to achieve the same statistical precision, the cost is significantly lower than the risks associated with data leakage. Large companies already employ this type of solution to protect sensitive user data.

In addition to differential privacy, researchers have developed a range of techniques to mitigate privacy violations, as we summarize in Table 2 (De Cristofaro, 2021; Liu et al., 2021). While Table 2 outlines specific technical methods, Table 3 complements them by presenting broader architectural and regulatory design principles. Rather than focusing on technical implementation, these principles offer foundational governance strategies and guidelines for structurally embedding privacy considerations into AI systems at an institutional and regulatory level.

In high-stakes domains involving large datasets, applying privacy-preserving techniques, such as those presented in Table 2, may require careful consideration of scalability and associated costs. While some of these techniques provide strong privacy guarantees, we must assess their computational overhead and potential impact on data utility according to the specific use case and context (Cummings et al., 2024; Near et al., 2024; Jin et al., 2023). Thus, selecting privacy-preserving methods becomes a strategic management decision, and organizations must classify data by sensitivity and apply complex solutions only where justified by risk.

3.2. Algorithm bias by bad data

Concerning the creation of biased ML models, there are two specific types of risks. The first issue with ML prediction models is that they may produce results correlated with race or gender, sometimes perpetuating social disparities. This kind of risk is typically associated with what we know as social bias (Olteanu et al., 2019). Social bias originates from existing prejudices within a given population, as captured in the data prior to the development of the ML model. It is important to distinguish this from what we call algorithmic bias. Algorithmic bias occurs when technical shortcomings during the model’s development cause systematic errors in outcomes. Although algorithmic bias can impact the accuracy of the models, we can often correct it by selecting a model of different complexity or by adjusting the relationship between attributes and target variables. In contrast, we cannot solve the issue of social bias simply by choosing a more accurate model. Even with optimal model accuracy, the underlying social biases in the data may still persist, reflecting broader societal prejudices that exist independent of the modeling process.

Table 3
Architectural and regulatory design principles for privacy.

Privacy by Design (Hustinx, 2010)	Embeds privacy into system architecture from the outset. Default settings and design choices should protect personal data without user intervention.
Data Minimization (Finck and Biega, 2021)	Limits data collection and use to what is strictly necessary for a specific purpose. Reduces exposure and risk by avoiding unnecessary data retention.
Informed Consent (Andreotta et al., 2022)	Critiques traditional consent in AI and big data. Proposes ethics oversight bodies and simplified, layered consent tools to enhance user understanding and institutional responsibility.
Privacy Impact Assessments (PIAs) (Wright and De Hert, 2012)	Assesses potential risks to individual privacy from data collection, use, and sharing within an organization.

An empirical example of social bias comes from COMPAS (Correctional Offender Management Profiling for Alternative Sanctions) and other similar systems, which are widely used to assist parole decisions⁴. In 1927, the new chairman of the Parole Board of Illinois noticed a gap in law enforcement efficiency compared to industry. Considering the public sentiment that the justice system favored criminals, he believed parole deserved a fair evaluation. He ordered a study by the University of Illinois, Northwestern University, and the University of Chicago. The study collected data on about 3000 Illinois parolees, categorizing them into groups such as “first offenders”, “occasional offenders”, “habitual offenders”, and “professional offenders”. Researchers investigated criminal, work, and residential histories, types of crimes, sentences, time served, and psychiatric diagnoses. Early findings showed clear patterns in parolee success and failure. For example, parolees with strong work records, high intelligence, and farming backgrounds who served less than a year had half the state average parole violation rate. Conversely, those from criminal backgrounds, whose prosecutors or judges had argued against leniency, and who served over five years, had twice the violation rate. The study suggested using this data to create summary sheets for parole decisions, arguing that rehabilitation could be predictable. Although only two states used such tools in 1970, by 2000, twenty-six states had adopted these models for parole decisions, making them standard practice. One of the most popular systems was COMPAS, which predicts whether a prisoner will reoffend within one to three years using factors such as age and criminal history. It also includes survey questions to identify individual needs, such as addiction or depression. A decade later, a journalist from a nonprofit news organization was surprised by how extensively the criminal justice system had adopted these tools without independent validation. In 2016, the journalist and her team published a report titled *Machine Bias*, which ran with the logline: “There is a software used across the country to predict future criminals. And it is biased against Blacks”.

The second type of risk relates to differential accuracy between groups. We may associate this type of risk with representation bias. Representation bias occurs when the input data is not representative of the relevant population, causing systematic errors in ML model predictions. For instance, [Klare et al. \(2012\)](#) investigate how demographic information, such as gender, race/ethnicity, and age-affects the performance of face recognition algorithms. They divide the dataset into cohorts based on gender (male, female), race/ethnicity (Black, White, Hispanic), and age group (18–30, 30–50, 50–70 years old). They find differences in classification accuracy across all three demographic categories. Specifically, they show that algorithms find female subjects more difficult to recognize than male subjects. Researchers attribute this performance gap to higher intraclass variability among females, possibly due to the use of cosmetics. They also show that Black subjects have the lowest recognition accuracy compared to White and Hispanic subjects. Training algorithms specifically on Black subjects improves recognition accuracy for this cohort, suggesting that unbalanced training data affects performance. In addition, younger subjects (18–30 years old) are more difficult to recognize than older age groups. Researchers attribute this to greater variability in younger individuals’ facial features over time, which may result from changes in appearance, style, and other dynamic factors. Training on age-specific cohorts improves performance within those groups. This example illustrates the difficulty models may face when used to identify individuals who have committed crimes. If the training database underrepresents certain groups or contains higher variability within specific populations, the model will perform less accurately for those individuals.

It is worth mentioning that, beyond the specific biases we discussed here, many other types of bias may occur in ML. Researchers provide a survey of these biases in [Olteanu et al. \(2019\)](#).

While typical solutions for the risk of differential accuracy among groups include augmenting the dataset or estimating models separately for different cohorts, developing ML models that avoid the perpetuation of biases present in the data remains a complex task. We generally develop ML models with the goal of maximizing some measure of accuracy, as discussed in [Section 2.3](#) (see, for instance, the estimation of the linear regression model or the logit model in that section). However, if the original data contains biases, the estimated models will likely reflect those biases.

One common approach to achieving fairness in ML is the removal of sensitive variables such as race or gender, a method known as fairness through unawareness. While this may initially seem effective, it often fails to eliminate bias. Other attributes may still allow the model to indirectly predict these protected variables, perpetuating discrimination. Moreover, excluding protected attributes, or attributes that can predict them, can prevent the model from reaching an optimal solution. Omitting this critical information can reduce model performance and fail to address the underlying sources of bias ([Kearns and Roth, 2019](#)). For example, in a model used to select students for a university, if race correlates with family income and, consequently, with access to quality education, simply ignoring race may perpetuate inequalities. Consider a hypothetical situation involving candidates of two races, “gold” and “silver”. “Gold” candidates, who are generally wealthier, may have had better access to education and therefore tend to perform better on admission tests. However, a “silver” candidate with a score of 7 may have more learning potential than a “gold” candidate with a score of 8. In this case, removing the race variable may lead to a less fair solution than adjusting admission thresholds for each race to balance access opportunities to the university.⁶

There are many techniques for mitigating the absence of fairness in ML ([Hort et al., 2024](#)). We can categorize them into three groups: pre-processing, in-processing, and post-processing. Pre-processing techniques modify the dataset before training to ensure equal representation of both protected and unprotected groups in the desired outcomes. In-processing approaches involve modifying the learning algorithm itself to incorporate fairness constraints or to augment the performance index during model estimation. These

⁴ The interested reader may find a complete description of the COMPAS in [Christian \(2020\)](#).

⁵ We say “supposedly” because, although some argue that COMPAS violated certain fairness metrics, it did not violate others. We will return to this point at the end of this section when we discuss the “impossibility theorem”.

⁶ This example is a slight variation of the one presented in [Kearns and Roth \(2019\)](#).

Table 4
Overview of bias mitigation techniques by stage.

Pre-processing techniques	
Remove Proxy Variables	Eliminates features strongly correlated with protected attributes. This method reduces direct discrimination risk but can impair model utility by discarding important information.
Dataset Augmentation (ADASYN)	Generates synthetic data for underrepresented groups to balance training distribution and improve fairness in model learning.
Fair Representation Learning	Learns new representations of the data that are informative for prediction but uncorrelated with protected attributes, ensuring group parity in the transformed feature space.
Disparate Impact	Modifies feature values to equalize the conditional distribution of features across groups, aiming to ensure fair treatment before model training.
Relabeling via Ranking	Reassigns labels to favor protected groups based on ranking scores, correcting imbalances while maintaining a ranking-consistent logic.
In-processing techniques	
Reweighting	
Assigns instance weights during training to balance class distributions across demographic groups and enforce fairness metrics.	
Fairness Constraints in Optimization	
Adds convex constraints (e.g., on covariance) to optimization to prevent decision boundary unfairness during training.	
Fairness Reductions via Cost-Sensitive Classification	
Transforms fair classification into a series of cost-sensitive classification problems guided by conditional moment constraints.	
Adversarial Debiasing	
Trains a predictor alongside an adversary that attempts to detect protected attributes; the predictor learns to mask those biases.	
Fairness Regularization	
Adds a fairness-oriented penalty term (e.g., mutual information with protected attributes) to the loss function.	
Bayesian Fairness	
Balances expected utility with fairness-aware Bayesian priors to achieve fair outcomes under uncertainty.	
Fair Ensemble Learning	
Combines multiple models, giving higher weight to those that perform better on fairness metrics across instances.	
Post-processing techniques	
Threshold Adjustment	Calibrates decision thresholds per group to satisfy fairness criteria like equal opportunity or equalized odds.
Transformation of Predicted Probabilities	Applies optimization-based score transformations to satisfy fairness constraints while preserving utility.
Reject Option Classification	Introduces a “reject zone” near decision boundaries to allow relabeling of uncertain predictions for improved fairness.

techniques ensure that the model optimizes not only for accuracy but also for fairness. Post-processing techniques apply adjustments to the model’s predictions after training. Their objective is to correct any biases in the predictions without changing the original dataset or the learning algorithm. [Table 4](#) summarizes key techniques used to mitigate algorithmic bias, categorized by the stage at which they intervene in the ML pipeline.

It is important to highlight a significant result in this field known as the impossibility theorem. This theorem states that no model can simultaneously satisfy multiple fairness criteria ([Kleinberg et al., 2016](#)). Furthermore, an important yet less explored issue concerns how different bias mitigation strategies affect individual predictions and whether they introduce arbitrariness into the debiasing process ([Goethals et al., 2024](#)). In particular, a major challenge with fairness interventions lies in their tendency to generate arbitrary outputs for individual samples.

Scalability significantly impacts bias mitigation techniques, particularly the post-processing approaches presented in [Table 4](#). Although these methods initially seem lightweight because they avoid retraining entire models, applying them at scale introduces substantial computational and operational challenges. These challenges can slow systems and impair responsiveness. In high-frequency decision environments, such as finance, healthcare, or criminal justice, post-processing methods often require solving complex optimization problems or applying score transformations on the fly, introducing latency that can degrade user experience and overall system performance ([Nandy et al., 2022](#)). Therefore, in large-scale, high-stakes settings, best practices include adopting pre-processing or in-processing solutions, which embed fairness directly into model training. While these approaches may require significant upfront computational investment, they yield fairness by construction and reduce ongoing operational complexity and computational costs in the long term.

[Table 5](#) summarizes architectural and regulatory design principles aimed at mitigating algorithmic bias arising from biased or unrepresentative data. Unlike the technical interventions summarized in [Table 4](#), these principles emphasize embedding ethical considerations directly into governance processes, data management policies, stakeholder engagement, and oversight mechanisms. By doing so, they help ensure that AI systems align with broader societal values and fairness objectives.

Table 5
Architectural and regulatory design principles for mitigating algorithmic bias from biased data.

Principle	Summary
Value-Sensitive Design (Simon et al., 2020)	Embeds human values into system design through conceptual, empirical, and technical investigations, ensuring systems align with ethical goals.
Representative Data Governance (Clemmensen and Kjærsgaard, 2022)	Promotes policies to ensure datasets are diverse and representative, reducing bias from unbalanced or skewed data.
Algorithmic Impact Assessments (Selbst, 2021)	Evaluates bias and impacts before deployment, encouraging documentation and mitigation of risks early in system design.
Bias Audits & Monitoring (Raji et al., 2020)	Involves periodic audits to detect and correct bias in data or outputs, ensuring ongoing fairness and accountability.
Transparency & Documentation (Geburu et al., 2021)	Creates structured documentation (e.g., datasheets, model cards) to disclose data and model limitations, aiding bias detection and oversight.
Inclusive Design & Stakeholder Engagement (Holstein et al., 2019)	Engages diverse teams and communities in system development to identify and reduce design biases.

3.3. Lack of interpretability and explainability

The risk of lacking *interpretability* and *explainability* in ML models arises when we build complex relationships between the features of observations and their associated predictions to maximize accuracy, but those relationships become difficult to understand. Interpretability refers to the ability to directly comprehend the model's structure and the relationships between inputs and outputs, while explainability involves using external methods to understand how inputs affect outputs.

A notable example highlighting the risk of lack of transparency was the controversial use of algorithmic grading by Ofqual, the UK's exams regulator ([Katwala, 2020](#)). In 2020, the COVID-19 pandemic led to the cancellation of the UK's final A-level exams, which are critical for university admissions. To determine student grades, Ofqual developed an algorithm based on three key factors: each school's historical grade distribution from the previous three years, individual student rankings within each subject provided by teachers, and students' prior exam results. When grades were announced, nearly 40 % of students received lower scores than those predicted by their teachers. This disproportionately affected students from less wealthy schools, as the algorithm prioritized historical school performance over individual student achievement. Consequently, talented students at historically underperforming schools received unfairly low grades. In contrast, students from private schools often benefited because such schools typically had smaller classes and consistently strong historical results. The algorithm placed greater trust in teacher assessments from these smaller, high-performing classes. As students faced results they could neither understand nor justify, widespread frustration erupted, severely undermining trust in both the algorithmic grading system and the broader educational framework.

It is important to note that only the simplest models, which tend to learn less complex relationships, such as the linear and logit models discussed in [Section 2.3](#), are inherently interpretable.⁷ However, we rarely choose these models because of their limited ability to capture complex patterns in the data. On the other hand, several advanced techniques are available to explain the results of more complex models. We summarize the most popular techniques in [Table 6](#), and readers can find a detailed review in [Molnar \(2020\)](#).

Note that making ML models explainable requires careful study to select the right approach and often involves costly additional computations. Explainability techniques, such as those in [Table 6](#), face distinct scalability challenges, especially when applied to highly complex models in large-scale or high-dimensional settings. Methods like Shapley values or SHAP often suffer significant performance degradation as model complexity and dataset size increase, making real-time explanations impractical in high-frequency environments ([Van den Broeck et al., 2022](#)). Permutation importance generally remains scalable and efficient, although computational costs can rise notably when applied to very large datasets or models with slow inference speeds. LIME offers another valuable alternative, particularly when we need explanations for a limited number of instances.

Note that selecting the best approach involves considering not only computational costs but also the architectural design principles that guide interpretability and explainability. We summarize key architectural principles in [Table 7](#). It is worth mentioning that, in general, we do not need detailed explanations for every decision. We should prioritize resources for critical applications such as medical diagnoses, credit assessments, or judicial rulings. In these exceptionally high-stakes scenarios, where the cost of error is significant, a recommended architectural principle is to adopt inherently interpretable models from the outset, sacrificing some predictive accuracy to ensure interpretability and reduce computational demands. This approach supports scalable explanations and helps minimize unnecessary system complexity.

⁷ Linear models and logit models are not the only interpretable models. For instance, decision trees are also interpretable because their simple structure mimics human decision-making processes. Their hierarchical structure lets us trace the decisions made at each node of the tree, making it straightforward to understand how we derive the final decision or classification. Additionally, decision trees clearly highlight the most important features used in making the decision.

Table 6
Interpretability and explainability techniques in ML.

Citation	Summary
Partial Dependence Plots (PDPs)	Visualize the average effect of a feature on the predicted outcome by marginalizing over the distribution of all other features. They are useful for global interpretation but can be misleading when features are correlated.
Individual Conditional Expectation (ICE) plots	Show the dependence of the predicted response on a feature for individual observations, providing a more granular view than PDPs and capturing heterogeneous effects.
Accumulated Local Effects (ALE) plots	Improve on PDPs by accounting for correlations between features. They compute feature effects locally using finite differences, avoiding unrealistic data combinations.
Permutation Importance (PI)	Assesses feature importance by measuring how model accuracy degrades when feature values are randomly shuffled, breaking their relationship with the target.
Grouped Permutation Importance (GPI)	Extends PI by evaluating the collective importance of variable groups, which is useful when assessing structured inputs such as grouped or hierarchical features.
Leave-One-Covariate-Out (LOCO)	Evaluates the marginal contribution of each feature by retraining the model without it and comparing performance, revealing individual feature influence but at higher computational cost.
Global Surrogate Models	Use interpretable models (e.g., decision trees) to approximate the predictions of complex models. They offer a global explanation but can misrepresent behavior if the fit is poor.
LIME	Explains individual predictions by fitting simple interpretable models locally around each prediction, revealing which features contribute most to a specific output.
Anchors	generate if-then rules (anchors) that explain model behavior with high-precision guarantees on a local decision boundary, offering intuitive and stable explanations.
Shapley values	fairly allocate feature contributions to predictions by computing the average marginal contribution of a feature across all possible subsets of features.
SHAP	unifies Shapley values with model-specific approximations for faster computation, preserving desirable theoretical properties and enabling local interpretability of complex models.

Table 7
Architectural and regulatory design principles for interpretability and explainability in ML.

Design Principle (Source)	Summary
Transparency by Design (Felzmann et al., 2020)	Embeds transparency into AI system design from the outset, guiding how decisions are made understandable at technical and organizational levels.
Right to Explanation (Kaminski, 2019)	Grants individuals a legal right to receive understandable reasons for impactful algorithmic decisions, fostering trust and accountability.
Human-in-the-Loop Governance (Jones, 2017)	Requires human oversight at key decision points, ensuring ethical alignment and preventing unchecked automation.
Procedural Regularity (Kroll et al., 2017)	Ensures AI decisions follow clear, consistent rules, as crucial for algorithmic accountability, highlighting tools such as transparency, auditing, cryptographic commitments ^a , zero-knowledge proofs ^b , and fair random choices ^c .
Interpretable Models (Transparent Design) (Rudin, 2019)	Favor using inherently interpretable models in high-stakes settings over opaque black-box systems.
Standardized Transparency Disclosures (Arnold et al., 2019)	Use documentation tools like model cards or factsheets to disclose system intent, performance, and limitations.
Multilayered Explanations Standard (Kaminski and Malgieri, 2020)	Defends the production of multi-layered explanations, including systemic explanations (overall system logic), group explanations (decisions affecting specific groups), and individual explanations (specific outcomes for individuals).

^a A cryptographic commitment allows an AI system to commit to a specific decision rule or process without revealing it publicly. Later, if accountability is required, the system can reveal the committed rule and prove noone has altered it.

^b A zero-knowledge proof allows an AI system to demonstrate that it followed a specific rule or made a decision correctly, without revealing the rule itself or the underlying data.

^c Fair random choices refer to mechanisms that ensure an AI system makes decisions based on genuinely random processes when appropriate, such as in tie-breaking or resource allocation, while allowing external verification that noone manipulated the randomness.

4. What a regulation should ask: principles of Ethical AI

In this section, we review the most relevant principles of ethical AI, which guide the kind of AI we aim to develop. These principles are crucial for several reasons. First, they help mitigate the risks associated with AI technologies. Second, they promote the responsible use of AI, ensuring that we distribute its benefits fairly across society. Third, they foster public trust in AI technologies, which is essential for their widespread adoption and integration into various aspects of daily life. Table 8 summarizes key principles of ethical AI. Each row highlights a fundamental ethical principle frequently discussed in the AI ethics literature (Jobin et al., 2019; Hoffmann-Riem, 2020).

While Table 8 presents the ethical principles together, it is essential to emphasize that these principles differ not only in content but also in the nature of the challenges they pose for regulation. Specifically, computer scientists have extensively studied the principles of privacy, justice and fairness, and transparency in the literature, and today there is a wide range of computational

Table 8
Summary of ethical principles for AI regulation.

Principle (Reference)	Explanation
Privacy	Safeguarding personal data against misuse, emphasizing informed consent and controlled data access.
Justice and Fairness	Ensuring equality by preventing biases and discriminatory impacts of AI algorithms, considering fairness metrics and balancing fairness with accuracy.
Transparency	Making AI decisions clear and understandable, focusing on interpretability, explainability, and documentation.
Responsibility	Holding developers and users accountable, emphasizing integrity, clear attribution of legal liability, and fostering a culture of ethical behavior.
Non-maleficence	Ensuring AI systems do no harm, prioritizing safety, security, and protection from misuse or harmful impacts.
Beneficence	Promoting positive impacts and societal well-being, encouraging beneficial AI applications while closely supervising those with negative impacts.
Freedom and Autonomy	Respecting and supporting individual autonomy in decision-making and data control, requiring clear consent and transparency.

techniques available to address them. The main barriers to implementation tend to be managerial or operational in nature, related to model scalability or trade-offs with predictive accuracy. As explored in [Section 3](#), researchers have developed concrete solutions such as differential privacy, fairness-aware learning algorithms, and post-hoc explainability tools. Regulatory strategies that promote these practices typically aim to standardize their use in critical applications. The main challenge lies in aligning legal requirements with evolving technical capabilities.

In contrast, the remaining principles, responsibility, non-maleficence, beneficence, and freedom and autonomy, are more normative and reflect deeper questions about how societies should use AI technologies. These principles are less amenable to purely technical fixes and require broader social, political, and legal engagement. For instance, responsibility raises issues of liability attribution and institutional accountability, while beneficence and non-maleficence compel us to define what constitutes social good or harm in contested domains. Similarly, freedom and autonomy extend beyond data protection to encompass concerns about manipulation, surveillance, and behavioral influence. Addressing these principles in regulation often demands public deliberation, ethical framing, and institutional design—rather than code alone. Therefore, a robust regulatory framework must not only mandate the implementation of technical solutions where they exist but also guide broader societal choices about which AI uses should be encouraged, constrained, or prohibited.

5. Strategies for regulating AI systems

This section explores the strategies used to regulate AI systems. Although various approaches to AI regulation exist, regulators have been considering three main strategies: *self-regulation*, discussed in [Section 5.1](#); *risk-based command-and-control regulation*, explored in [Section 5.2](#), and *innovation-driven regulation*, presented in [Section 5.3](#).

Furthermore, we conclude this section by presenting two important aspects that apply across different regulatory strategies: the *sector-based approach*, discussed in [Section 5.4](#), and the *strict versus subjective liability* approach, presented in [Section 5.5](#).

5.1. Self-regulation of AI

Self-regulation means that industries, sectors, or individual companies create and manage their own voluntary rules or standards, such as some of the principles presented in [Section 4](#), without direct government control ([Bartle and Vass, 2007](#)). This approach allows companies to update their standards quickly to match new technologies or market conditions. Companies usually adopt these voluntary rules due to market pressure, the need to maintain consumer trust, the desire to compete effectively, or the intention to avoid stricter government regulation in the future. Industry groups or peer organizations typically oversee compliance, applying sanctions such as fines or public criticism. However, these penalties are generally less severe than those imposed by governments.

Self-regulation offers important advantages. It allows fast responses to technological changes and reduces the burden on government resources, giving authorities more freedom to focus elsewhere. Yet, self-regulation also presents important disadvantages. Companies that enforce their own rules may encounter conflicts of interest, as profit incentives can compete with strict compliance. Although many companies have effectively tried to balance ethical standards and commercial goals, public trust in corporate self-regulation tends to be lower due to these perceived risks. Additionally, different companies or sectors may enforce standards unevenly, leading to inconsistent protection for consumers.

5.2. Risk-based command-and-control regulation of AI

Command-and-control regulation explicitly defines which actions are illegal by clearly specifying the required or forbidden behaviors related to technology management and use ([Stewart, 1991](#)). A common contemporary example of this type of regulation is the risk-based approach, where regulators focus on identifying and reducing risks that can cause significant harm to the public or the environment. When regulating AI systems specifically, regulators need to conduct thorough risk assessments of current AI applications ([Guihot et al., 2017](#)). This process involves defining acceptable risk levels, consulting with industry stakeholders, and gathering detailed sector-specific information. Collecting comprehensive data helps regulators understand risks more effectively, set clear priorities, and allocate resources efficiently.

Command-and-control regulation has clear advantages. It provides explicit and detailed legal standards that reduce ambiguity and simplify compliance. However, this regulatory style faces several challenges when applied to AI. One major issue is that AI-related risks change frequently, depending on factors such as the type of AI application, the generality of its use, the specific AI model, and the potential severity of negative outcomes (Guihot et al., 2017). Another significant challenge is the speed at which AI technology evolves. Traditional command-and-control rules usually cannot keep pace with rapid technological advancements, potentially limiting their effectiveness for emerging technologies such as AI. Moreover, overly prescriptive regulations, those mandating the use of specific technologies or methods may sometimes constrain innovation by reducing the flexibility companies have to explore alternative solutions. This effect may be particularly pronounced for smaller companies and startups, which often have fewer resources to meet stringent regulatory requirements. As a result, strict regulations might unintentionally create barriers to market entry and growth, possibly discouraging some innovative initiatives or new entrants altogether.⁸

5.3. Innovation-driven regulation

Innovation-driven regulation focuses on flexible frameworks designed to keep pace with new technologies. Instead of strict rules that companies must follow exactly, innovation-driven regulation relies on broad guidelines, known as soft laws, which describe the outcomes society seeks, such as the principles listed in Section 4, while allowing flexibility in how companies achieve these goals (Marchant and Allenby, 2017). In addition, this approach uses practical tools like regulatory sandboxes, where companies can test new products safely under supervision before fully introducing them into the market (Burd, 2021). This method helps regulators better understand new technologies, promoting innovation while protecting the public interest.

Companies following innovation-driven regulation receive oversight both internally and externally (Gutierrez et al., 2020). Internal oversight comes from boards of directors, ethics committees, or ombudsmen who help ensure that companies follow ethical standards. External supervision involves multiple actors, including supply chains (with customers demanding responsible practices), certifications, audits, insurers (who set conditions in insurance policies), funding agencies (that select projects based on ethical and safety standards), and government procurement policies. Governments can also influence companies directly by requiring adherence to ethical standards as a condition for public contracts or funding.

Innovation-driven regulation brings several advantages. It is highly flexible, adapting quickly as technology evolves, and actively promotes innovation through structured experimentation. By involving multiple stakeholders, such as regulators, companies, and independent organizations, it builds trust and credibility in the regulatory process. It also lowers initial compliance costs, particularly benefiting small companies and startups. However, innovation-driven regulation can be challenging to implement. It requires significant resources for monitoring and regular adjustments. Its flexible nature may lead to inconsistencies across sectors or regions, creating confusion for businesses. The close relationship between regulators and industry participants increases the risk of regulations unfairly favoring certain companies or sectors. Finally, the dynamic and evolving rules may cause uncertainty, making compliance difficult, especially for smaller firms with fewer resources.

5.4. The sectorial versus the non-sectorial perspective

The sectorial approach adapts regulation specifically to individual sectors or industries, reflecting their unique characteristics, risks, and needs. This targeted approach ensures that regulations directly address the specific issues, technologies, and business practices within a given sector. Conversely, regulators design non-sectorial approaches to apply across multiple sectors or industries. Despite their broad applicability, these regulations remain flexible enough to accommodate a wide range of activities and business models, providing a fundamental layer of compliance.

When choosing between sectorial and non-sectorial approaches, several considerations come into play (Bacache-Beauvallet and Perrot, 2017). More complex and highly specialized sectors may benefit more from sectorial regulations due to the specific risks and technologies involved. Furthermore, in fast-evolving fields, non-sectorial approaches offer the flexibility needed to apply regulatory principles broadly without requiring frequent, sector-specific updates.

When regulating AI, the non-sectorial approach may face difficulties in defining general regulatory principles. This challenge stems from the inherent difficulty of clearly defining what constitutes AI, as discussed in Section 2.1, and from the wide variety of AI applications across domains. In particular, the non-sectorial command-and-control regulations face notable challenges in accommodating heterogeneous industry contexts due to their rigidity and specificity. Such regulations might impose uniform technological or procedural standards across diverse sectors, potentially ignoring sector-specific risks, innovation patterns, and technological maturity.

5.5. Strict versus subjective liability approach

A crucial issue in AI regulation is deciding how regulators should assign responsibility when AI systems cause problems or harm (Kheterpal, 2024). One common option is strict liability, where companies take full responsibility for harm caused by their systems,

⁸ For example, Klapper et al. (2006) show that costly regulations discourage new companies, especially in industries that normally attract many new entrants. Such regulations also force new companies to start at larger sizes and slow the growth of existing firms. Additionally, Palagashvili and Suarez (2024) demonstrate, using data from the United States and Canada, that highly regulated sectors have fewer new startups and higher rates of closure for small companies.

Table 9
Challenges in regulating AI.

Challenge (Reference)	Explanation
Translation of ethical principles into regulatory frameworks	Difficulty in concretely embedding ethical values and standards into practical and enforceable AI regulations.
Automation fallacies	Misunderstandings in AI debates, including expecting perfection, unfair AI-human comparisons, and oversimplified ‘adopt or ban’ mentality.
Research and Development (R&D) characteristics	AI development is often discreet, diffuse, discrete, and opaque, complicating monitoring and control.
Government absence in AI development	Predominantly private-driven AI development leaves governments less informed and capable of identifying key regulatory needs.
Information asymmetry	Regulators lack essential technical knowledge and understanding compared to private AI developers, hindering effective oversight.
Coordination across regulatory bodies	Effective regulation requires interdisciplinary coordination, creating complexity and risks of inconsistent or incomplete standards.
Agency capture	Regulators becoming overly influenced by AI industry interests due to frequent personnel exchanges between regulatory bodies and industry.
Limited enforcement mechanisms and jurisdiction shopping	Major AI firms leveraging resources and mobility to circumvent regulations, evade supervision, or seek favorable regulatory jurisdictions.

regardless of their intentions or preventive actions. Regulators often apply this strict approach to high-risk technologies to ensure that companies remain cautious and committed to safety.

However, an alternative option is subjective liability. Under subjective liability, regulators consider the company’s intentions, preventive measures, and compliance efforts before assigning responsibility. This method may apply, for instance, to companies that obtain formal certification for their AI systems, demonstrating that they followed accepted safety standards and practices (Gutierrez et al., 2020). Subjective liability may foster innovation and encourage companies to seek certification and comply carefully with safety standards, while also promoting fairness by taking into account their efforts to avoid harm.

6. Challenges to regulate AI systems

Regulating AI systems involves unique challenges that differ from those associated with other public risks, due to AI’s inherent characteristics (Scherer, 2015; Guihot et al., 2017; Hoffmann-Riem, 2020; Lobel, 2023). We summarize these challenges in Table 9.

7. Final thoughts

In this paper, we provided a comprehensive review of AI regulation, exploring key aspects such as AI definitions, applications, inherent risks, guiding principles for AI development, and the challenges involved in regulating it. We differentiated our approach by integrating perspectives from computer science, law, and economics to offer a unified view of AI regulation. This multidisciplinary perspective supports a rigorous understanding of the complexities involved, making our work valuable to a broad audience.

We should not frame AI regulation as a battle of extremes. Instead, it should represent a careful effort to balance individual and societal safety with the need to foster innovation. Navigating these trade-offs requires clarity and a nuanced understanding of several critical issues. Defining AI is inherently difficult, as no universally accepted definition exists. Relying on vague or flawed definitions introduces legal uncertainty and can undermine the effectiveness of regulatory frameworks. AI applications vary widely across sectors, technologies, models, levels of autonomy, automation capabilities, and data types. This diversity suggests that a uniform regulatory approach may not only prove ineffective but also hinder the development of beneficial AI applications. Nevertheless, despite this complexity, there is broad consensus on the essential qualities AI systems should uphold, such as privacy protection, justice and fairness, transparency, responsibility, non-maleficence, beneficence, and the preservation of individual freedoms. Regulation should explicitly aim to safeguard these principles. Given the specific risks posed by AI, regulatory efforts must remain precise rather than overly broad. Generalist regulations often fail to address unique or nuanced challenges, leaving critical vulnerabilities unaddressed.

Computer scientists have extensively studied AI risks and developed a wide range of practical solutions. Ignoring these established technical tools when designing regulations would be a serious oversight that weakens regulatory effectiveness. At the same time, we should not presume that all companies lack ethical commitments or disregard their reputations. Nor should we assume that all companies act ethically by default. Regulation must reflect this variability and create incentives that encourage positive behavior. While AI regulation is essential, we must also avoid assuming that stricter rules are always better. Excessively rigid regulations can stifle innovation, slowing technological progress and economic growth in a rapidly evolving field. Recognizing the complexity and variability of AI applications, different sectors will inevitably require distinct regulatory approaches. It is unrealistic to expect a small group of legislators or regulators to fully understand the range of risks and benefits across all domains. Effective regulation requires inclusive processes that draw on expertise from government, industry, academia, and computer science.

In legal contexts, AI system certification offers a valuable mechanism for distinguishing between strict and subjective liability. Certification helps clarify legal responsibility in disputes and encourages companies to follow established safety standards. Regulatory sandboxes also serve as an important tool. They allow developers to test AI systems under controlled conditions, identify

risks, and refine technologies before full deployment. Ensuring broad and easy access to these environments can improve AI system quality, proactively address risks, and create a safer path for innovation.

Finally, to assist regulators in navigating the complex landscape of AI regulation, several key questions deserve careful consideration. An initial and critical question is whether those responsible for formulating AI regulations possess sufficient expertise and a deep understanding of AI technologies, their capabilities, and their associated complexities. Equally important is how regulators incorporate public input into the regulatory process. Regulators must examine whether effective mechanisms exist to ensure that a broad and diverse range of stakeholder perspectives are adequately considered and integrated into regulatory frameworks. They should also evaluate how their strategies address the inherently challenging, and potentially flawed, definitions of AI. This includes considering the level of uncertainty these definitions introduce and how that uncertainty may affect the clarity and effectiveness of regulatory efforts. Another essential consideration is the alignment of regulatory policies with established ethical principles for AI development, such as fairness, transparency, privacy, and accountability. Regulators must actively ensure that their policies uphold and reinforce these standards.

The economic impact of AI regulations also demands close attention. Regulators should assess whether their policies foster or hinder innovation, economic growth, and productivity. Regulatory frameworks must provide clear guidelines without unnecessarily constraining technological progress or economic vitality. In addition, regulators should consider how their policies influence broader societal well-being. They must evaluate whether their efforts genuinely enhance public welfare by addressing societal concerns and promoting inclusive social benefits.

Given the rapid evolution of AI technologies, flexibility in regulatory frameworks is also crucial. Regulators must assess whether their rules can adapt to future advancements and how easily they can update those rules as new challenges and opportunities emerge. Furthermore, regulators need to strike a careful balance to avoid both overregulation, which can stifle innovation, and underregulation, which can leave major risks unmanaged. A balanced approach is essential to ensure both safety and progress. Lastly, ensuring compliance is vital.

Regulators must confirm that clear, robust, and enforceable mechanisms are in place to promote adherence to standards. Strong compliance mechanisms foster accountability, reinforce trust, and support the responsible deployment of AI technologies across sectors.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We sincerely thank the Ministry of Management and Innovation in Public Services, especially the Secretariat of Digital Government, for their financial support and valuable input on this project. We also appreciate the insightful discussions with Antonio Guimaraes from the Banco Central do Brasil (BACEN), Denise Gontijo from the Ministério do Planejamento e Orçamento (MPO), Eduardo Paracência from the Escola Nacional de Administração Pública (ENAP), Guilherme Jose from the Banco Central do Brasil (BACEN), Hugo Souto from Secretaria de Governo Digital (SGD), Ieda Chaves from Secretaria de Governo Digital (SGD), Thaciana Cerqueira from Secretaria de Governo Digital (SGD), and Wildi de Oliveira from Secretaria de Governo Digital. Additionally, DOC is grateful to CNPQ for their partial financial support (grant 304706/2023-0).

References

- Agarwal, S., Mishra, S., 2021. *Responsible AI Implementing Ethical and Unbiased Algorithms*. Springer, London.
- Andreotta, A.J., Kirkham, N., Rizzi, M., 2022. Ai, big data, and the future of consent. *Ai Soc.* 37, 1715–1728. <https://doi.org/10.1007/s00146-021-01262-5>
- Arnold, M., Bellamy, R.K., Hind, M., Houde, S., Mehta, S., Mojsilović, A., Nair, R., Ramamurthy, K.N., Olteanu, A., Piorkowski, D., et al., 2019. Factsheets: increasing trust in ai services through supplier's declarations of conformity. *IBM J. Res. Dev.* 63, 1–6. <https://doi.org/10.1147/JRD.2019.2942288>
- Bacache-Beauvallet, M., Perrot, A., 2017. Economic regulation: which sectors to regulate and how? *Notes du Cons. danalyse Econ.* 44, 1–12.
- Bartle, I., Vass, P., 2007. Self-regulation within the regulatory state: towards a new regulatory paradigm? *Public Adm.* 85, 885–905. <https://doi.org/10.1111/j.1467-9299.2007.00684.x>
- Burd, J.T., 2021. Regulatory sandboxes for safety assurance of autonomous vehicles. *Univ. Pa. J. Law Public Aff.* 7, 194.
- Christian, B., 2020. *The Alignment Problem: Machine Learning and Human Values*. WW Norton & Company.
- Clemmens, L.H., Kjærsgaard, R.D., 2022. Data Representativity for Machine Learning and AI Systems. *arXiv preprint arXiv:2203.04706*. <https://arxiv.org/abs/2203.04706>.
- Cummings, R., Desfontaines, D., Evans, D., Geambasu, R., Huang, Y., Jagielski, M., Kairouz, P., Kamath, G., Oh, S., Ohrimenko, O., et al., 2024. Advancing differential privacy: where we are now and future directions for real-world deployment. *Harv. Data Sci. Rev.* 6, 1–123. <https://doi.org/10.1162/99608f92.d3197524>
- De Cristofaro, E., 2021. A critical overview of privacy in machine learning. *IEEE Secur. Priv.* 19, 19–27.
- Dwork, C., 2006. Differential privacy. *International colloquium on automata, languages, and programming*. Springer, pp. 1–12. https://doi.org/10.1007/11787006_1
- Felzmann, H., Fosch-Villaronga, E., Lutz, C., Tamò-Larrieux, A., 2020. Towards transparency by design for artificial intelligence. *Sci. Eng. Ethics* 26, 3333–3361. <https://doi.org/10.1007/s11948-020-00276-4>
- Finck, M., Biega, A.J., 2021. Reviving purpose limitation and data minimisation in data-driven systems. *Technol. Regul.* 2021, 44–61. <https://doi.org/10.71265/z7r0t122>. <https://techreg.org/article/view/10986>.
- Fredrikson, M., Lantz, E., Jha, S., Lin, S., Page, D., Ristenpart, T., 2014. Privacy in pharmacogenetics: an end-to-end case study of personalized warfarin dosing. In: *Proceedings of the 23rd USENIX Conference on Security Symposium*, USENIX Association, USA. 17–32. 10.5555/2671225.26712.

- Gebru, T., Morgenstern, J., Vecchione, B., Vaughan, J.W., Wallach, H., Iii, H.D., Crawford, K., 2021. Datasheets for datasets. *Commun. ACM* 64, 86–92. <https://doi.org/10.1145/3458723>
- Goethals, S., Calders, T., Martens, D., 2024. Beyond accuracy-fairness: Stop evaluating bias mitigation methods solely on between-group metrics. *arXiv preprint arXiv:2401.13391*.
- Guihot, M., Matthew, A.F., Suzor, N.P., 2017. Nudging robots: innovative solutions to regulate artificial intelligence. *Vanderbilt J. Entertain. Technol. Law* 20, 385. <https://doi.org/10.2139/ssrn.3775271>
- Gutierrez, C.I., Marchant, G., Tournas, L., 2020. Lessons for artificial intelligence from historical uses of soft law governance. *Jurimetrics* 61, 133. <https://doi.org/10.1109/ACCESS.2019.2928979>
- Hoffmann-Riem, W., 2020. Artificial intelligence as a challenge for law and regulation. *Regul. Artif. Intell.* 1–29. https://doi.org/10.1007/978-3-030-32361-5_1
- Holstein, K., Vaughan III, J.W., Daumé, H., Dudik, M., Wallach, H., 2019. Improving fairness in machine learning systems: what do industry practitioners need? *CHI '19: Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. pp. 1–16. <https://doi.org/10.1145/3290605.3300830>
- Hort, M., Chen, Z., Zhang, J.M., Harman, M., Sarro, F., 2024. Bias mitigation for machine learning classifiers: a comprehensive survey. *ACM J. Responsible Comput.* 1, 1–52. <https://doi.org/10.1145/3631326>
- Hustinx, P., 2010. Privacy by design: delivering the promises. *Identit. Inf. Soc.* 3, 253–255. <https://doi.org/10.1007/s12394-010-0061-z>
- Jin, W., Yao, Y., Han, S., Gu, J., Joe-Wong, C., Ravi, S., Avestimehr, S., He, C., 2023. Fedml-he: An Efficient Homomorphic-encryption-based Privacy-preserving Federated Learning System. *arXiv preprint arXiv:2303.10837*.
- Jobin, A., Ienca, M., Vayena, E., 2019. The global landscape of AI ethics guidelines. *Nat. Mach. Intell.* 1, 389–399. <https://doi.org/10.1038/s42256-019-0088-2>
- Jones, M.L., 2017. The right to a human in the loop: Political constructions of computer automation and personhood. *Soc. Stud. Sci.* 47, 216–239. <https://doi.org/10.1177/0306312717699716>
- Kaminski, M.E., 2019. The right to explanation, explained. *Berkeley Technol. Law J.* 34, 189–218. <https://doi.org/10.15779/Z38TD9N83H>
- Kaminski, M.E., Malgieri, G., 2020. Algorithmic impact assessments under the gdpr: Producing multi-layered explanations. *Int. Data Priv. Law* 10, 19–28. <https://doi.org/10.1093/idpl/izp011>
- Katwala, A., 2020. An Algorithm Determined Uk Students' Grades. *Chaos Ensued. Wired.* August.
- Kearns, M., Roth, A., 2019. *The Ethical Algorithm: the Science of Socially Aware Algorithm Design*. Oxford University Press.
- Kheterpal, J., 2024. Choosing the Appropriate Standard of Liability for Artificial Intelligence-related Harms.
- Klapper, L., Laeven, L., Rajan, R., 2006. Entry regulation as a barrier to entrepreneurship. *J. Financ. Econ.* 82, 591–629. <https://doi.org/10.1016/j.jfineco.2005.09.006>
- Klare, B.F., Burge, M.J., Klontz, J.C., Bruegge, R.W.V., Jain, A.K., 2012. Face recognition performance: Role of demographic information. *IEEE Trans. Inf. Forensics Secur.* 7, 1789–1801. <https://doi.org/10.1109/TIFS.2012.2214212>
- Kleinberg, J., Mullainathan, S., Raghavan, M., 2016. Inherent Trade-offs in the Fair Determination of Risk Scores. *arXiv preprint arXiv:1609.05807*.
- Kroll, J.A., 2018. The fallacy of inscrutability. *Philos. Trans. R. Soc. A Math. Phys. Eng. Sci.* 376, 20180084. <https://doi.org/10.1098/rsta.2018.0084>
- Kroll, J.A., Huey, J., Barocas, S., Felten, E.W., Reidenberg, J.R., Robinson, D.G., Yu, H., 2017. Accountable algorithms. *Univ. Pa. Law Rev.* 165, 633–705.
- Liu, B., Ding, M., Shaham, S., Rahayu, W., Farokhi, F., Lin, Z., 2021. When machine learning meets privacy: a survey and outlook. *ACM Comput. Surv.* 54, 1–36.
- Lobel, O., 2023. The law of AI for good. *Fla. Law Rev.* 75, 1073.
- Marchant, G.E., Allenby, B., 2017. Soft law: new tools for governing emerging technologies. *Bull. At. Sci.* 73, 108–114. <https://doi.org/10.1080/00963402.2017.1288447>
- Martinez, R., 2018. Artificial intelligence: distinguishing between types & definitions. *Nev. Law J.* 19, 1015.
- McCarthy, J., et al., 2007. What is Artificial Intelligence.
- Molnar, C., 2020. *Interpretable Machine Learning*. Lulu. com.
- Nandy, P., Diccio, C., Venugopalan, D., Logan, H., Basu, K., ElKaroui, N., 2022. Achieving fairness via post-processing in web-scale recommender systems, In: *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency*, 715–725. <https://doi.org/10.1145/3531146.3533136>
- Narayanan, A., Shmatikov, V., 2006. How to break the anonymity of the netflix prize dataset. *Cornell University: Cryptography and Security*.
- Narayanan, A., Shmatikov, V., 2008. Robust de-anonymization of large sparse datasets, In: *2008 IEEE Symposium on Security and Privacy (sp 2008)*, IEEE. 111–125. <https://doi.org/10.1109/SP.2008.33>
- Near, J., Darais, D., Durkee, M., 2024. Scalability Challenges in Privacy-Preserving Federated Learning. *Technical Report*.
- Olteanu, A., Castillo, C., Diaz, F., Kiciman, E., 2019. Social data: biases, methodological pitfalls, and ethical boundaries. *Front. big data* 2, 13. <https://doi.org/10.3389/fdata.2019.00013>
- Palagashvili, L., Suarez, P.A., 2024. What is the relationship between industry-specific regulation and technology startups? *Economic Affairs* 10.1111/ecaf.12674.
- Raji, I.D., Smart, A., White, R.N., Mitchell, M., Gebru, T., Hutchinson, B., Smith-Loud, J., Theron, D., Barnes, P., 2020. Closing the AI accountability gap: defining an end-to-end framework for internal algorithmic auditing. *FAT* '20: Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*. pp. 33–44. <https://doi.org/10.1145/3351095.3372873>
- Rich, E., 1983. *Artificial intelligence*. McGraw-Hill, Inc.
- Rudin, C., 2019. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nat. Mach. Intell.* 1, 206–215. <https://doi.org/10.1038/s42256-019-0048-x>
- Ruschmeier, H., 2023. AI as a challenge for legal regulation-the scope of application of the artificial intelligence act proposal (Era Forum). *Springer*, pp. 361–376. <https://doi.org/10.1007/s12027-022-00725-6>. (Era Forum).
- Russell, S.J., Norvig, P., 2020. *Artificial Intelligence: A Modern Approach*. Pearson.
- Scherer, M.U., 2015. Regulating artificial intelligence systems: risks, challenges, competencies, and strategies. *Harv. J. Law Technol.* 29, 353.
- Selbst, A.D., 2021. An institutional view of algorithmic impact assessments. *Harv. J. Law Technol.* 35, 117.
- Simon, J., Wong, P.H., Rieder, G., 2020. Algorithmic bias and the value sensitive design approach. *Internet Policy Rev.* 9, 1–16. <https://doi.org/10.14763/2020.4.1534>
- Stewart, R.B., 1991. Models for environmental regulation: central planning versus market-based approaches. *Boston Coll. Environ. Aff. Law Rev.* 19, 547.
- Sweeney, L., 2000. Uniqueness of simple demographics in the us population. *LIDAP-WP4*, 2000.
- Van den Broeck, G., Lykov, A., Schleich, M., Suciu, D., 2022. On the tractability of shap explanations. *J. Artif. Intell. Res.* 74, 851–886. <https://doi.org/10.1613/jair.1.13283>
- Wright, D., De Hert, P., 2012. Introduction to privacy impact assessment. *Privacy impact assessment*. Springer, pp. 3–32. https://doi.org/10.1007/978-94-007-2543-0_1
- Zech, H., 2021. Liability for AI: Public Policy Considerations. *ERA forum*. Springer, pp. 147–158. <https://doi.org/10.1007/s12027-020-00648-0>